Examiner only

SECTION B

Answer **all** *questions in the spaces provided.*

7. *(a)* Melting temperatures vary down groups and across periods.

(i) Explain why chlorine is a gas but iodine is a solid at room temperature and pressure. [3]

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(ii) Explain why sodium has a lower melting temperature than aluminium. [1]

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(iii) Explain why silicon has a higher melting temperature than phosphorus. [1]

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(b) State and explain how you would expect the first ionisation energy of nitrogen to compare with the first ionisation energy of oxygen. [2]

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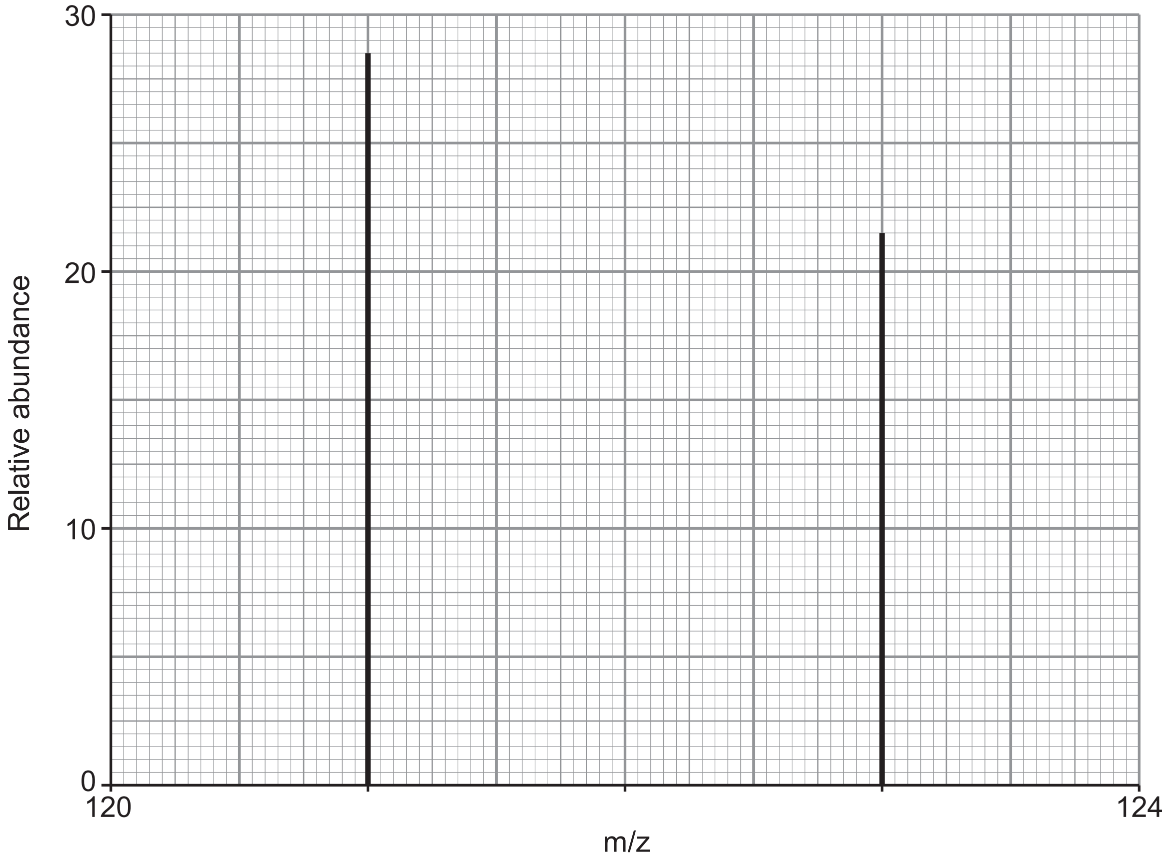
(c) Antimony is in Group 5. Its mass spectrum shows that it has two stable isotopes.



Calculate the relative atomic mass of antimony. You **must** show your working. [2]

Relative atomic mass = ...

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(d) Although radiation from radioisotopes is harmful to health many beneficial uses of radioactivity have been found.

The table below gives some information about four radioactive isotopes.

Isotope	Radiation emitted	Half-life
^{90}Sr	β	28 years
^{99}Tc	γ	6 hours
^{210}At	α	8.1 hours
^{228}Th	α	1.9 years

Use **all** the data given to choose which isotope is the most suitable to use as a tracer in medicine. Explain your answer. [3]

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